

Krish Lakshmi Narayanan

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Technical Skills

Languages: Python, SQL, R, SAS, Go, TypeScript/JavaScript

Machine Learning: Supervised Learning, Classification, Regression, Clustering, Feature Engineering, Neural Networks, PyTorch, TensorFlow, XGBoost, LightGBM, Scikit-learn

Generative AI: LLMs, LLM Evaluation, Prompt Engineering, LLM-as-a-Judge, OpenAI, Anthropic, Ollama, MCP

Data Engineering & MLOps: ML Data Pipelines, ETL, PySpark, Apache Spark, Databricks, Airflow, dbt, Fivetran, AWS S3, AWS Athena, PostgreSQL, Snowflake, BigQuery

Software & AI Engineering: REST APIs, distributed systems, microservices, SQL/NoSQL Databases, LLM-assisted software development

Experimentation & Statistics: A/B tests, Bayesian & frequentist methodologies, hypothesis testing, causal inference, Sampling

Professional Experience

Senior Analyst – Advanced Analytics & Optimization – CIBC – Toronto, Canada – (August 2025 – Present)

- Trained and productionized an **XGBoost** model achieving **0.86 AUC** across **500,000+** employee records, engineering behavioral signals to surface compliance anomalies, reducing manual review by **30%** and identifying high-risk cases **2x** faster than rule-based approaches
- Engineered **ML-ready ETL pipelines** processing **3B+** records using **Python, PySpark** and **Databricks**, decreasing end-to-end data latency by **35%** through optimized partitioning and parallel execution at multi-terabyte scale
- Architected production systems serving **10,000+ users** across distributed infrastructure, integrating **Python** services with cloud-native workflows to sustain high availability and eliminate single points of failure

Data Intern – 1Password – Toronto, Canada – (May 2024- August 2024)

- Developed and evaluated **LightGBM** and **logistic regression** machine learning models using **precision/recall** across user segments, applying **k-means clustering** for cohort analysis and increasing website engagement by **5%** through targeted funnel optimization
- Constructed production **ETL pipelines** ingesting **2B+** clickstream and **Google Analytics** records into **AWS S3** and **Athena**, maintaining **100% data integrity** across downstream ML experimentation and analytics workflows
- Stood up an experimentation framework and executed **5+ A/B tests** using Bayesian and frequentist methodologies, boosting experimentation throughput by **40%** and informing high-impact product launches across activation and retention funnels
- Translated complex multi-source analyses into **Looker dashboards** tracking conversion, activation, and user lifecycle metrics for senior stakeholders, accelerating decision speed by **25%**

Data Analyst – Super.com – Toronto, Canada - (January 2023 – April 2023)

- Developed pricing and revenue optimization models using **regression techniques** across **1M+ transaction** records, surfacing actionable spend patterns that contributed to **5%+ revenue growth** in key e-commerce segments
- Designed and executed **5+ A/B experiments**—sample size calculations, experiment design, and post-hoc statistical analysis via Bayesian and frequentist methodologies—generating customer behavior insights that produced a **5% net revenue lift** in key e-commerce segments
- Automated **15+ production datasets** and reporting workflows using **Airflow, FiveTran** and **dbt**, raising data reliability by **45%** and enabling scalable analytics infrastructure that freed analyst capacity for higher-leverage insight work

Small Business Data Analyst – Scotiabank – Toronto, Canada - (January 2022 – August 2022)

- Pioneered **statistical and regression-based** scoring models—linear regression—across **600+ advisors** to evaluate training certification, performance, and behavioral risk, directly informing compensation, promotion, and workforce planning decisions
- Applied predictive modeling techniques—**classification** and **probability-of-event modeling**—across **100,000+** customer and advisor records to assess churn and onboarding quality, combining approaches to sharpen retention strategy for financial products

Projects

LLM Evaluation & Prompt-Experimentation Platform – Python (FastAPI, PyMC, PyTorch/PEFT), Ollama, React/TypeScript, PostgreSQL

- Built a full-stack platform — along with **pe CLI**, and an **MCP judge tool** — that evaluates LLM prompts and models as **A/B experiment** arms
- Implemented paired significance testing (**bootstrap**, **Wilcoxon signed-rank**, **Holm-Bonferroni**) and a **PyMC** Bayesian equivalence test to compare arms statistically instead of by win-rate
- Calibrated an **LLM-as-judge** against hand-labeled data (Cohen's κ) and **QLoRA**-fine-tuned Qwen3-8B to a significant quality win (**90% vs 80% accuracy**, corrected $p=0.031$); **350+** automated tests and CI

TensorBuild – Visual ML Pipeline Builder – Python (FastAPI), React/Vite, Rust (Tauri), scikit-learn, PyTorch

- Built a cross-platform desktop app for visually building **ML pipelines**, preprocessing, model, and evaluation nodes into a **DAG**
- Designed a plugin **node system** (manifest + module) powering **31 node types** — **sklearn, PyTorch/CNN, evaluation**
- Set up **CI/CD** via GitHub Actions, backed by **170** automated engine tests, to package cross-platform desktop installers

QuantTrade – Live Demo – Python (FastAPI, SQLAlchemy, scikit-learn), React, PostgreSQL, Celery/Redis

- Built and deployed a **full-stack** platform for strategy backtesting via visual rules or **sandboxed custom Python/ML code**
- Designed an **async Celery/Redis** task architecture so multi-minute backtests never block the API
- Implemented walk-forward evaluation and **AST-sandboxed** execution, validated by **240+** automated tests and CI

Education

Honors Bachelor of Mathematics (Co-op) – University of Waterloo

Double Major: Computational Mathematics & Statistics | **Minor:** Economics

- **Dean's Honors** in Computational Mathematics and Statistics; **Term Honors distinction** in 8 academic terms
- **Coursework:** Data Structures & Algorithms, Neural Networks, Advanced Regression, Classification